

Magnetic Resonance Imaging Principles

BM 6090 & 3090 Course – Jan-Apr 2024

***Jaladhar Neelavalli, PhD.,
Dept. of of Biomedical Eng.
IIT Hyderabad***

Brief History

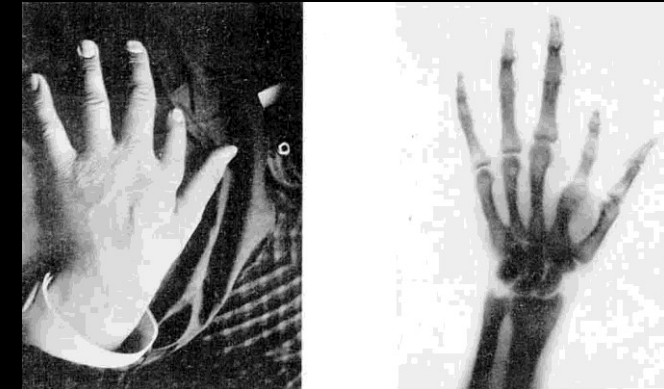
X - Rays



Wilhelm Roentgen
discovered in
November 1895



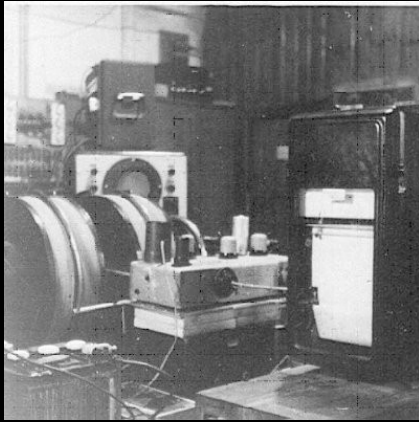
First X- ray image of
WR's wife's hand
December 1895



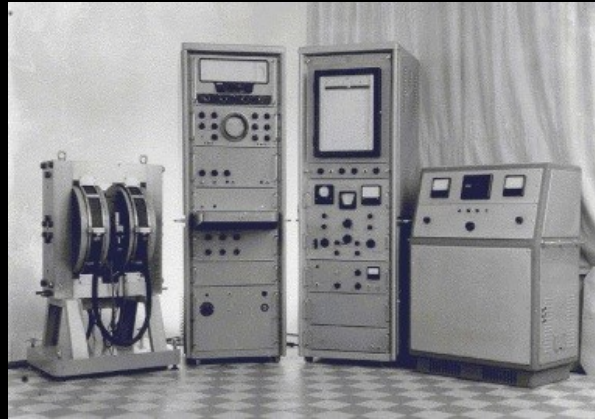
First medical use of X-ray
in diagnostics
Mid 1896

Brief History

NMR / MRI



Isidor Rabi first
described NMR
1938



1st commercial NMR
Spectrometers
1951



1st human MRI scanner
1977

Overview: The MRI system



Overview

Complexity → Versatility

Several ways to generate contrast between tissues!!!!!!

Intrinsic:

- spin density
(water content)
- T1/T2/T2*/T1ρ
- Chemical shift
- Magnetic Susceptibility
- Temperature
- Macromolecular content
- ...

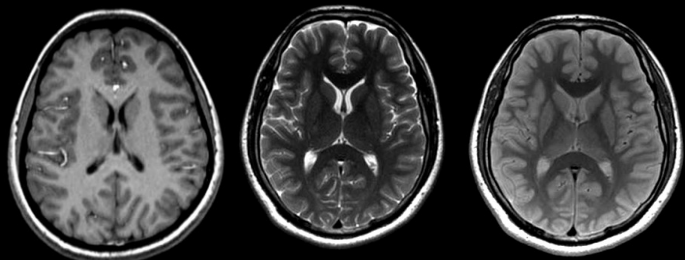
Physiological:

- Blood flow
- Perfusion
- Blood oxygenation
- Tissue elasticity
- Diffusion
- ...

Modes of Imaging:

- 2D/3D
- Steady state
- Dynamic Imaging
- Selective excitation /suppression
- Functional / interactive
- ...

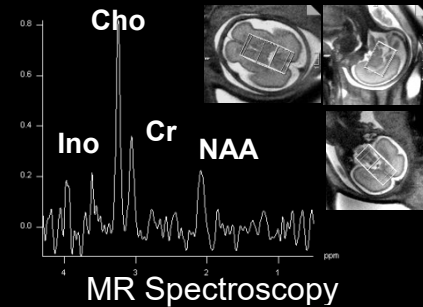
Overview



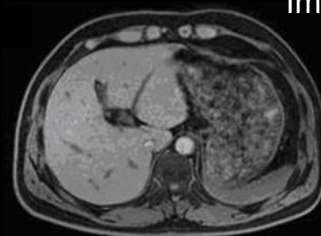
T1

T2

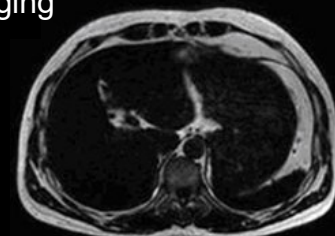
Water content
image



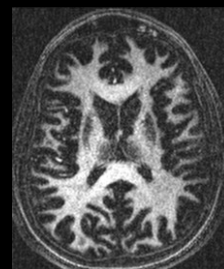
Tissue selective
Imaging



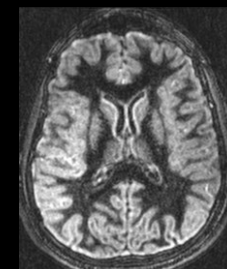
Water image



Fat image



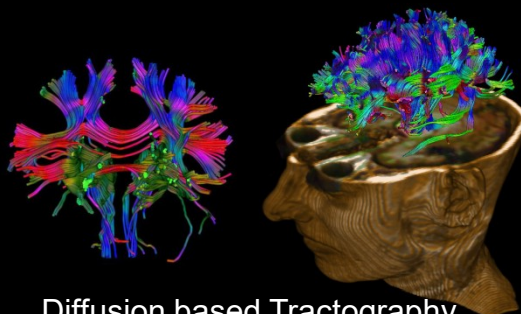
WM image



GM image

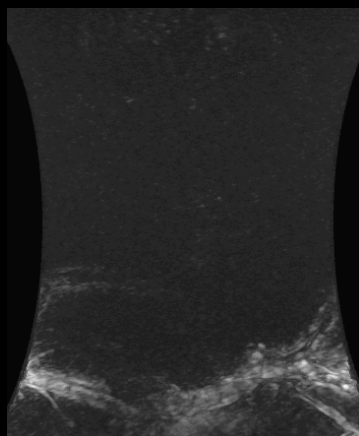


MR angiography (non-contrast)

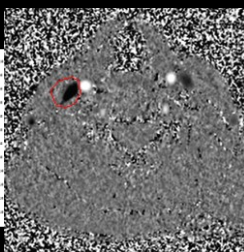
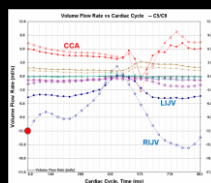


Diffusion based Tractography

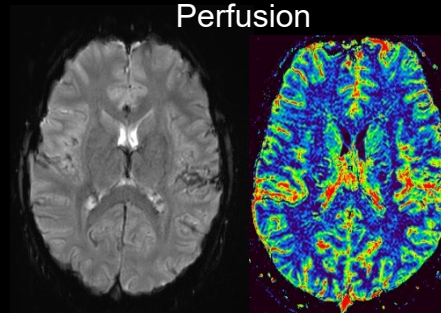
Dynamic Imaging



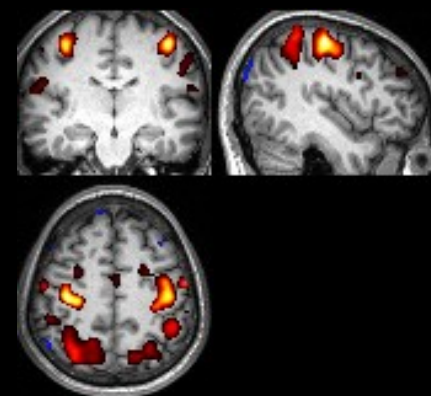
Cardio / cerebrovascular function



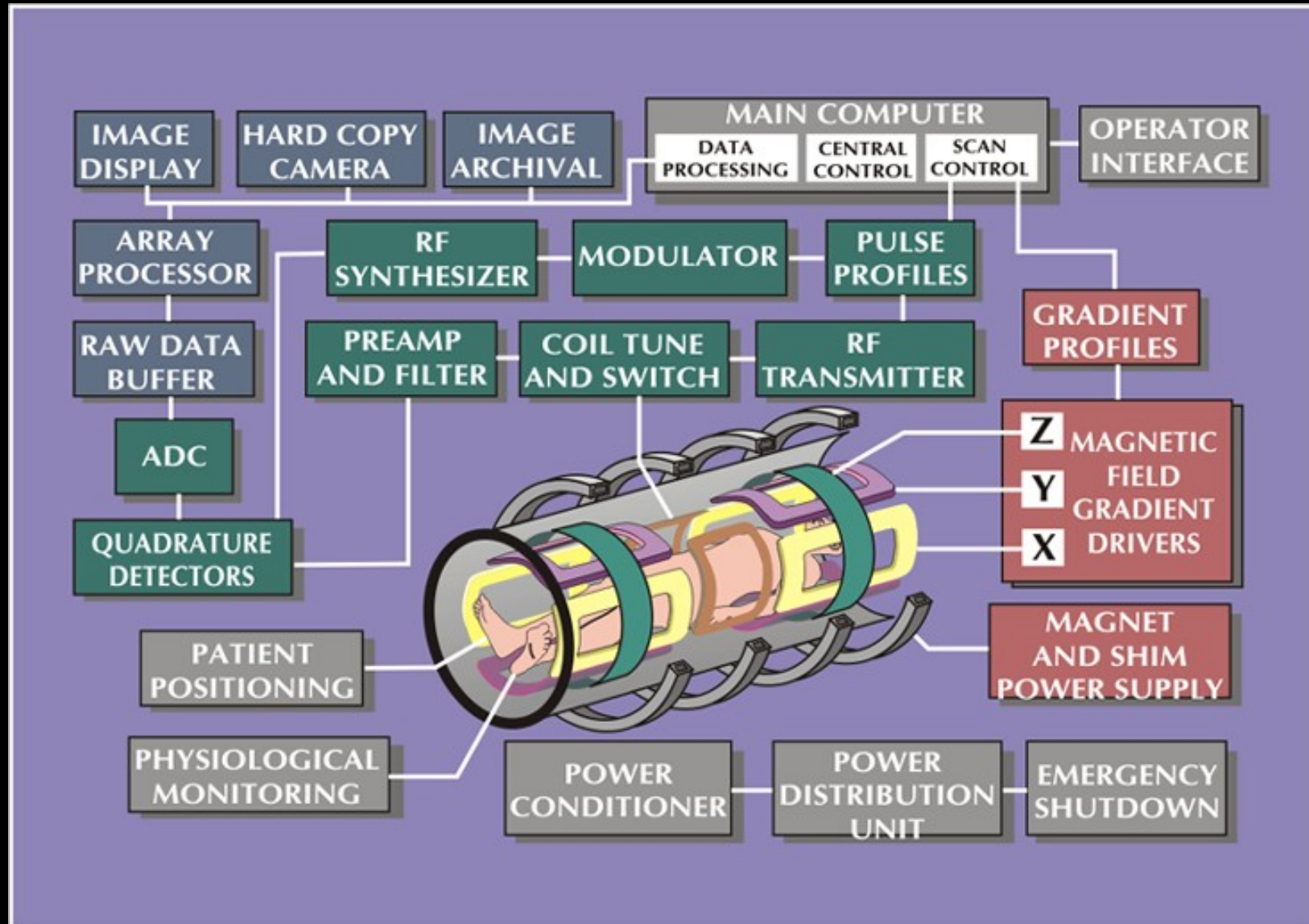
Perfusion



fMRI



Overview: The MRI system



The MRI system

- Magnet
- Gradient sub system
- RF (radio frequency) sub system
- Computer sub system
- (Patient monitoring system)

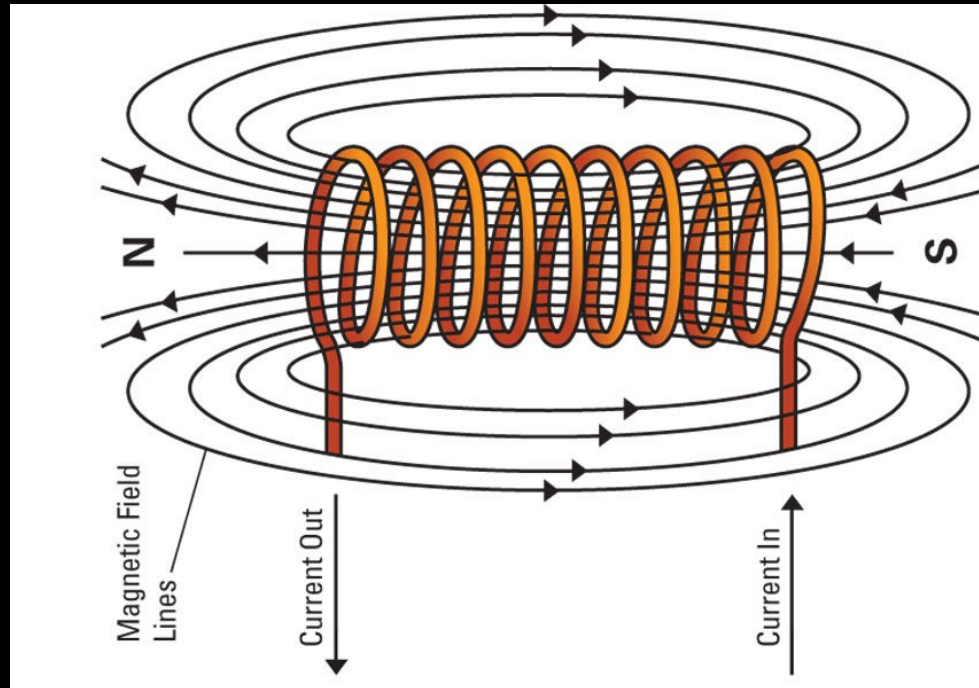
Overview: The MRI system

Magnet - To produce a uniform static magnetic field B_0 over a volume of interest for imaging

Earth's Magnetic field strength

5×10^{-5} Tesla
(or 0.5 Gauss)

1 Gauss – 10^{-4} Tesla

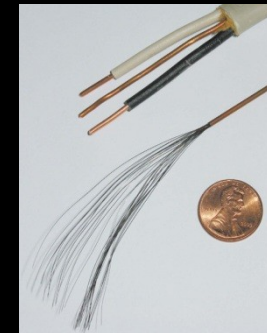


Typical clinical MRI field strength

1.5 Tesla
(15000 Gauss)

The Magnet (B_0)

- Superconductive magnet
 - Uses superconductive coils carrying very strong currents ($\sim 700\text{A}$) to generate the B_0
 - Requires very low temperature to maintain superconductive state (NbTi: 9K; Coolant: Helium, 4.2K or -269°C)
- Pros
 - Produce very strong field ($\geq 1.5\text{T}$)
 - B_0 remains very steady
 - Easier to achieve uniformity via shimming
- Cons
 - Expensive to manufacture and maintain
 - Potential hazards when quench



NbTi alloy
wire

Overview: The MRI system

Magnet with superconducting coils generating high field strengths

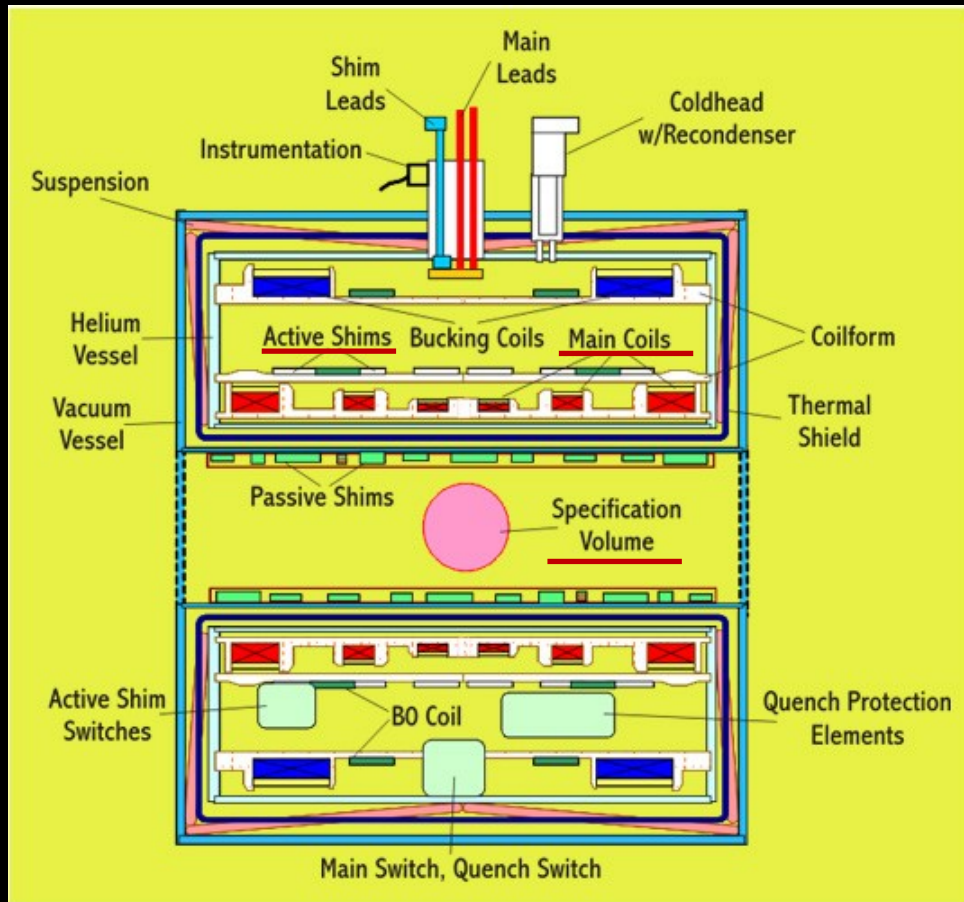


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Overview: The MRI system

Magnet with superconducting coils generating high field strengths

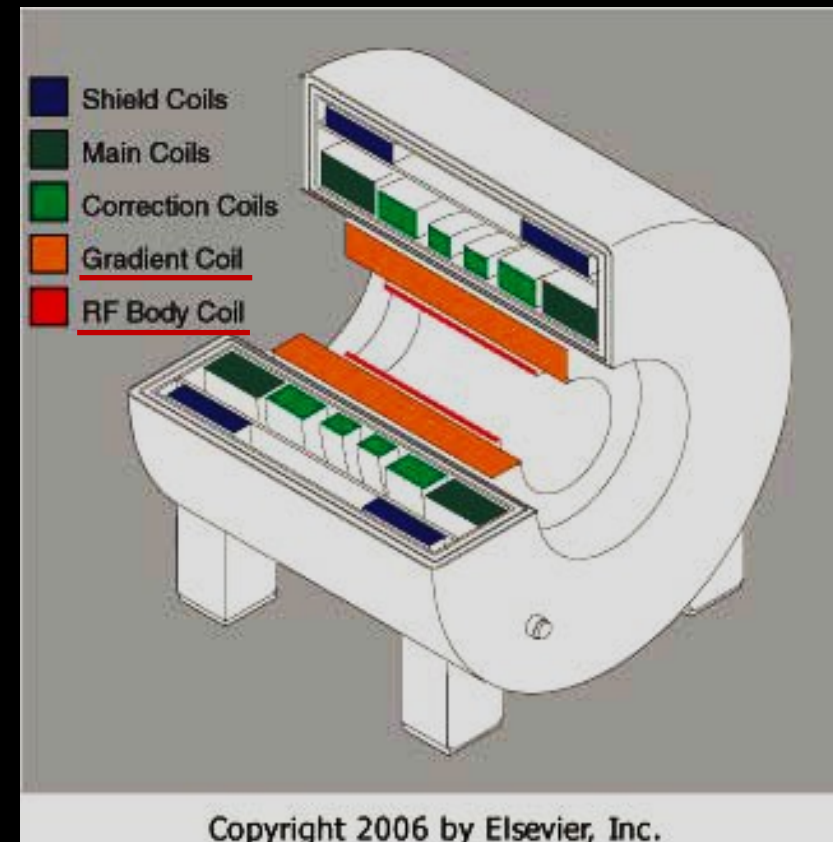
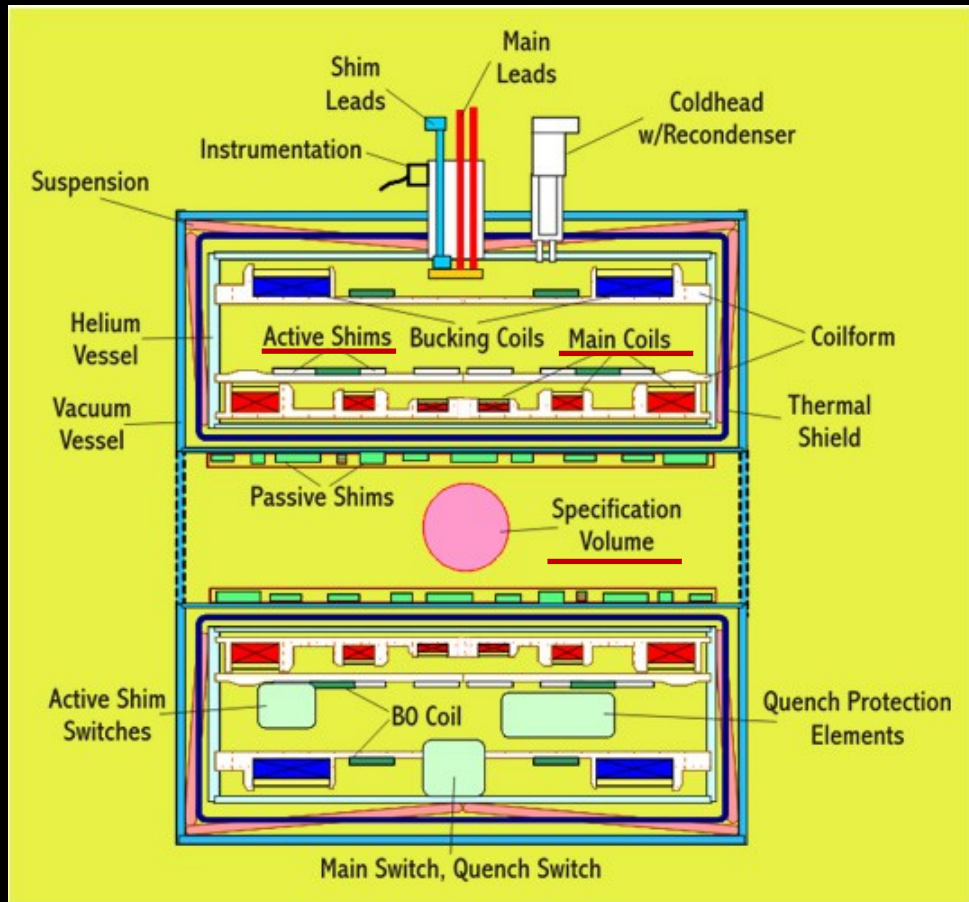


- DSV – Typical size is about 40 to 50 cms in the center of the bore
- Shim coils – Used to fine tune the uniformity of B_0 field

Typical magnetic field uniformity: 0.3×10^{-6} - 5×10^{-6} Tesla (or 0.3 to 5ppm)

Overview: The MRI system

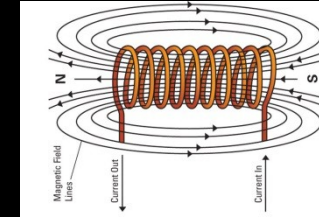
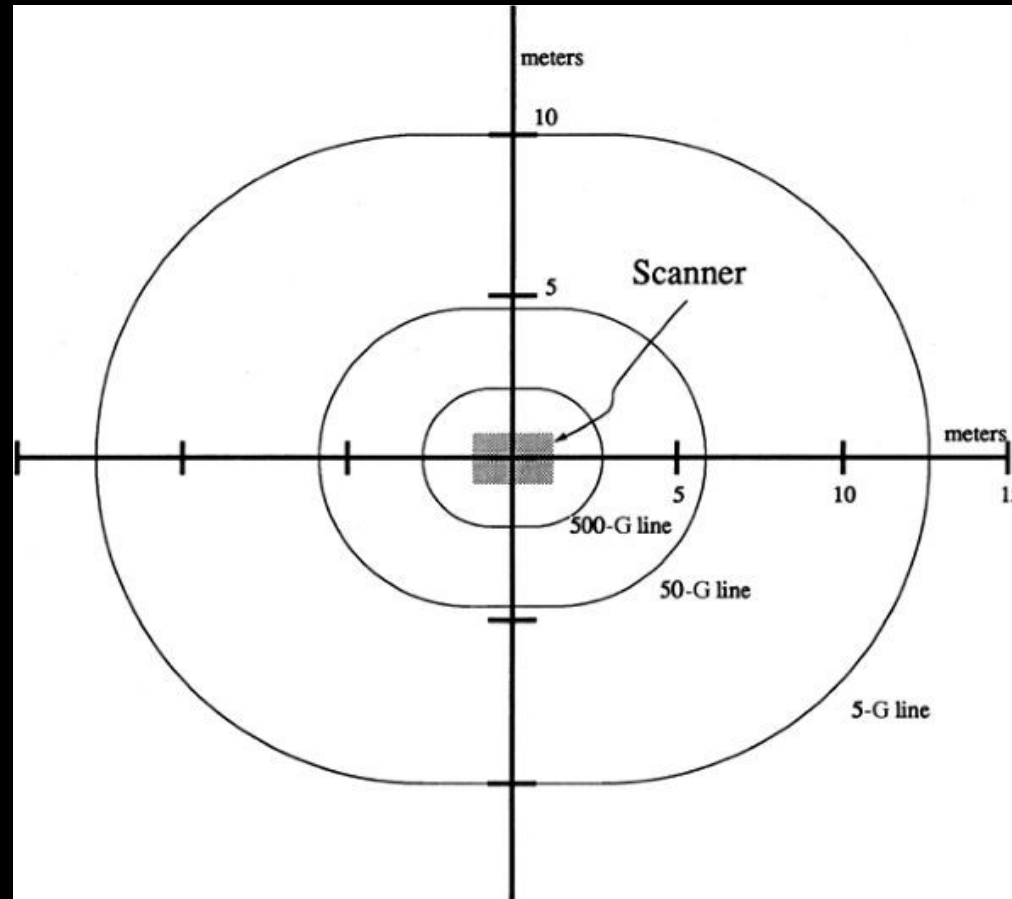
Magnet with superconducting coils generating high field strengths



Typical magnetic field uniformity: 0.3×10^{-6} - 5×10^{-6} Tesla (or 0.3 to 5ppm)

Overview: The MRI system

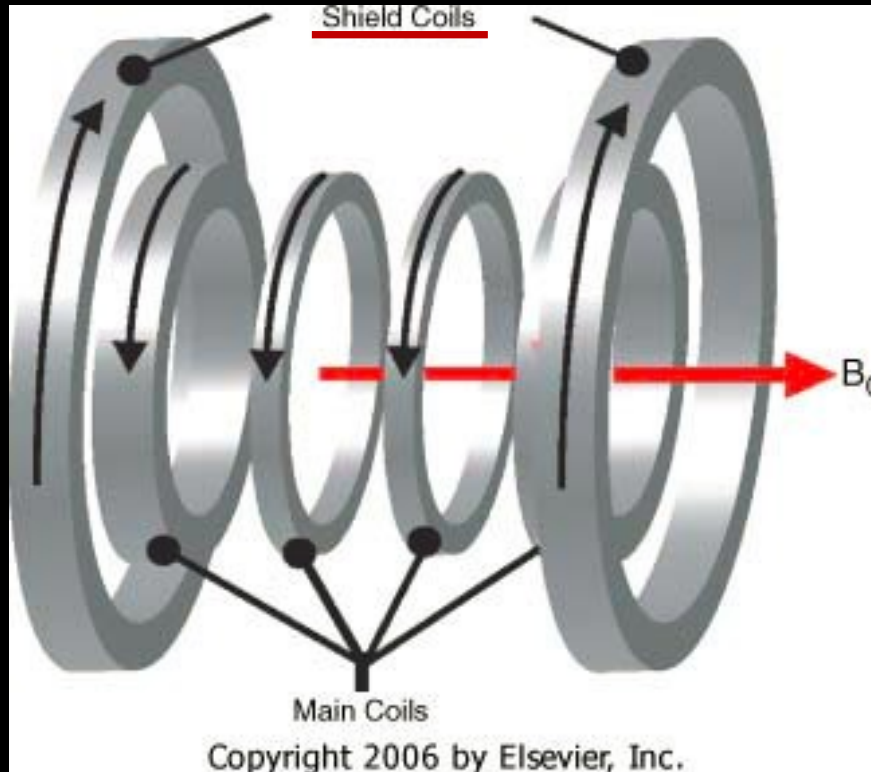
Magnet – Fringe Field



Strong magnets – large spatial extent of Fringe Fields

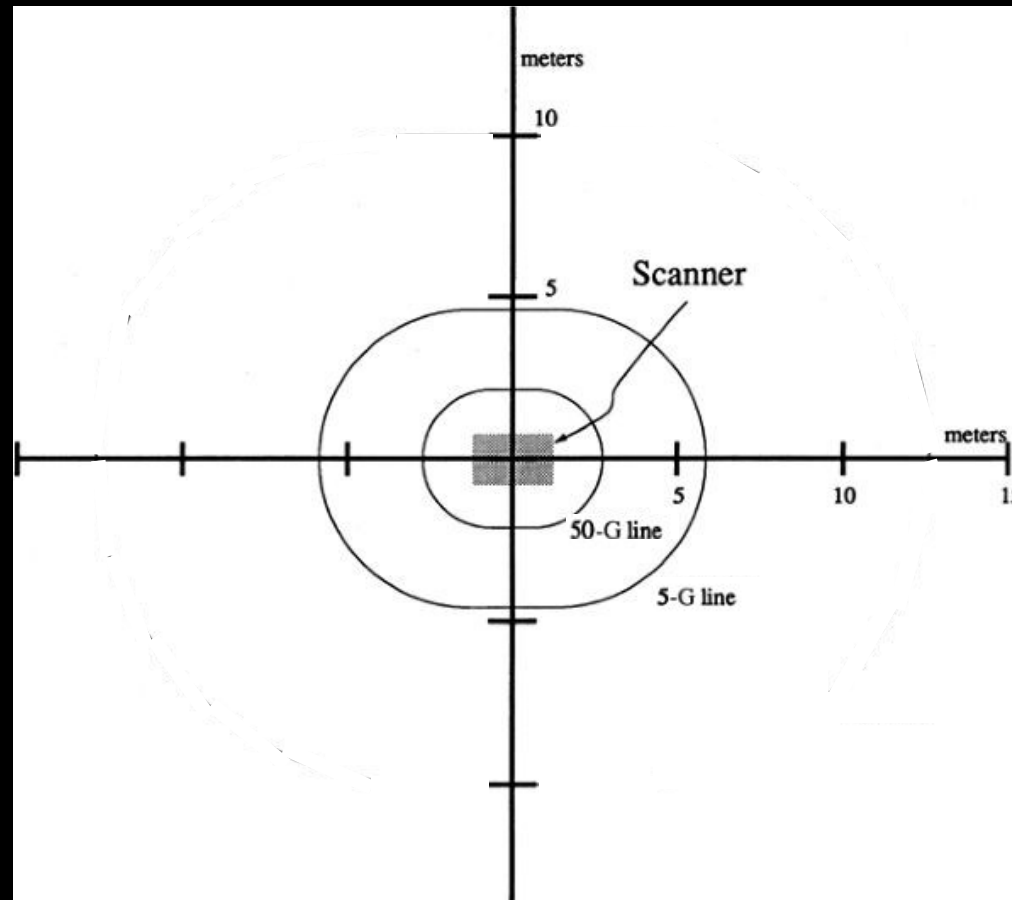
Overview: The MRI system

Reducing the spatial extent of fringe fields – Active Shielding



Overview: The MRI system

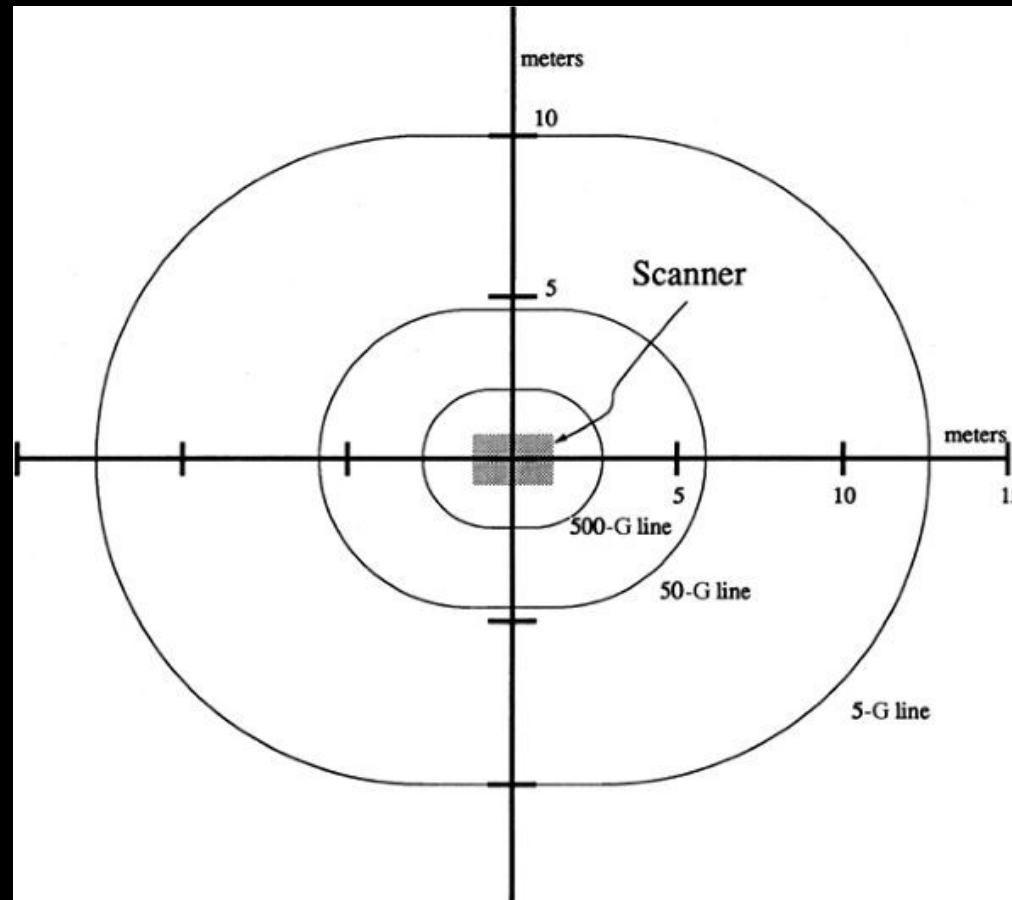
Magnet – Fringe Field



Active Shielding – Significantly reduces the extent of Fringe Fields

Overview: The MRI system

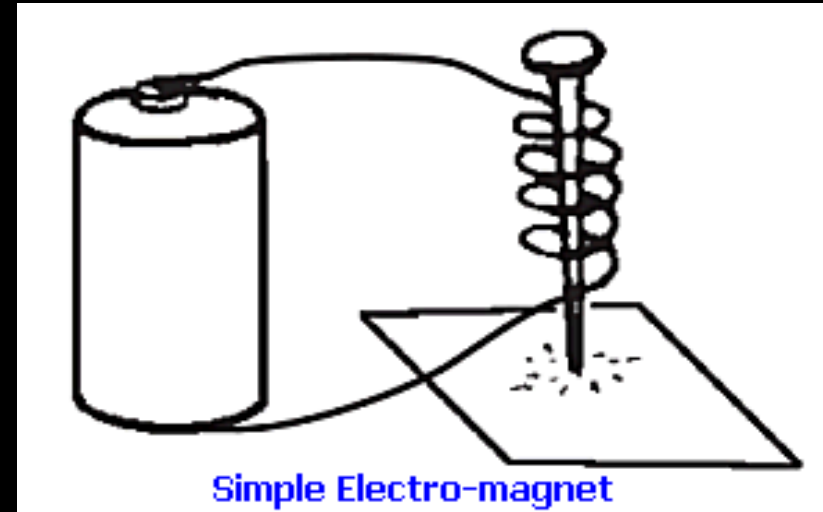
Magnet – Fringe Field



Strong magnets – large spatial extent of Fringe Fields

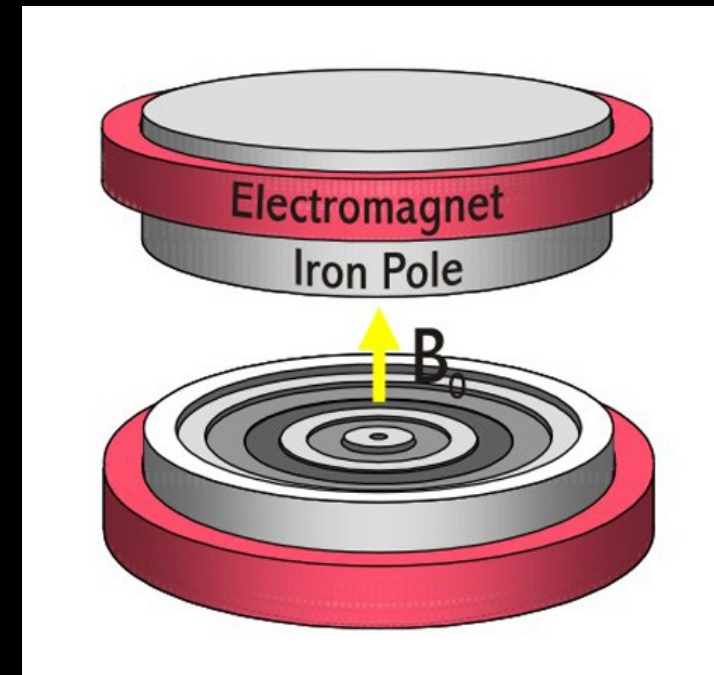
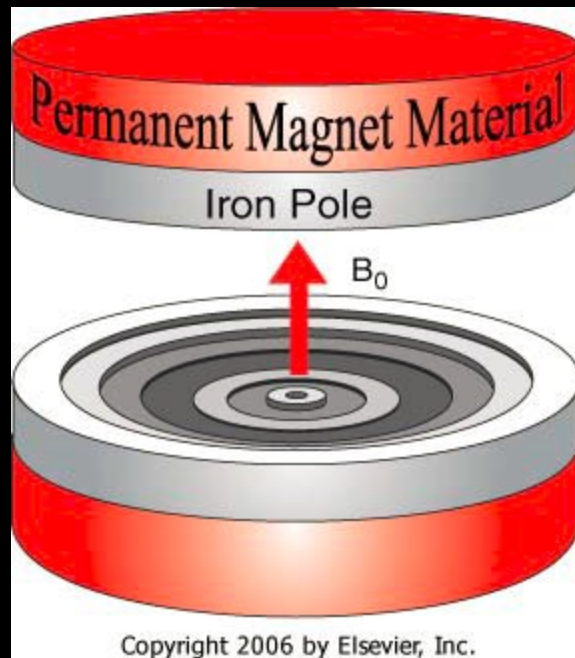
Other Magnet types

- Permanent and resistive magnets



Other Magnet types

- Permanent and resistive magnets



Other Magnet types – Permanent magnets

- Permanent magnets
 - Built from custom-shaped blocks of bar magnets
 - Usually with open design for scanner ('C' shaped)
- Pros
 - Economic
 - Open space, more comfortable (?)
- Cons
 - Restricted to low field ($<1\text{T}$, usually $0.2\sim0.45\text{T}$)
 - Limited field uniformity, and difficult to do shimming
 - Susceptible to temperature change
 - Eddy current effects
 - Bulky for installation



Other Magnet types – Resistive magnets

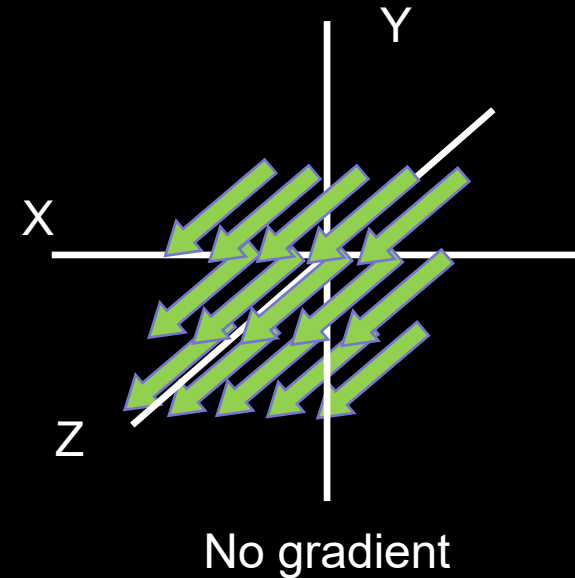
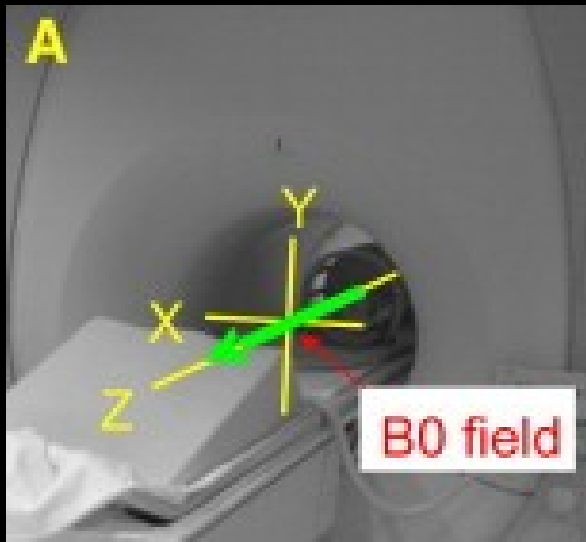
- Resistive magnet
 - Using normal copper wire, operate in room temperature
 - Magnetic field generated from the electromagnet mechanism
- Pros
 - Easy to design, build and routine maintain
 - Can be design to be vertical
- Cons
 - Consumes lots of electricity and creates lots of heat
 - Poor field stability due to heating
 - Low field strength
- Typically used for custom applications such as imaging the limbs etc.



Overview: The MRI system

Magnetic Field Gradient sub system

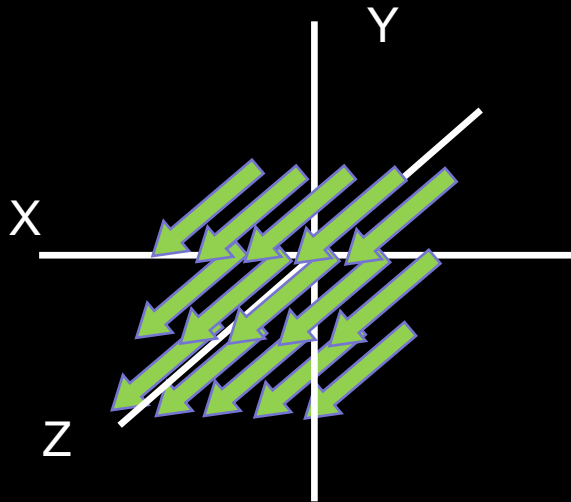
MFG simply varies the, otherwise uniform magnetic field along z direction, linearly as a function of x, y and z directions



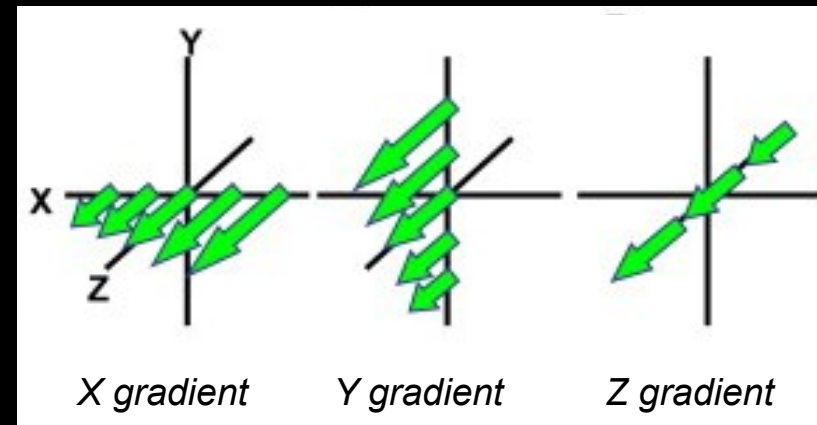
Overview: The MRI system

Magnetic Field Gradient sub system

MFG simply varies the, otherwise uniform magnetic field along z direction, linearly as a function of x, y and z directions



No gradient

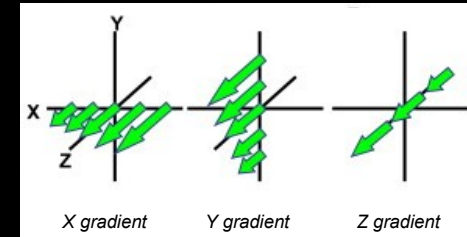
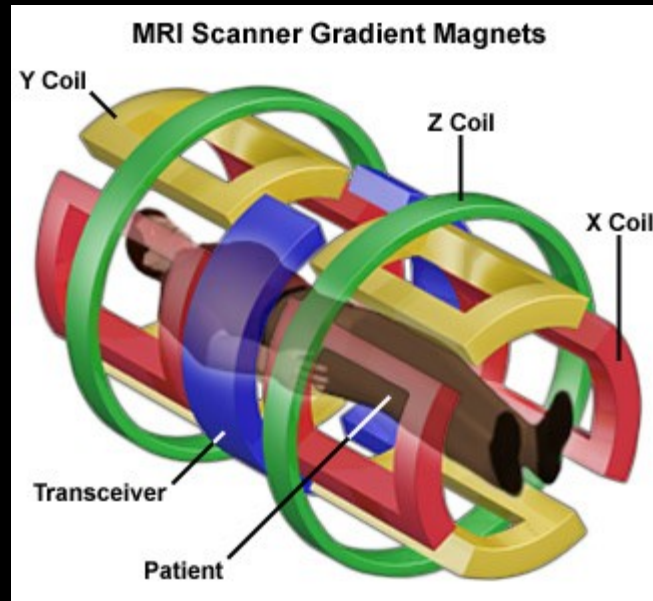


Linear field change as a function of space, for example (x)

$$B_0 = B_0 + G_x X$$

Overview: The MRI system

Magnetic Field Gradient sub system - To produce precisely defined spatial (x, y, z directions) variation in the otherwise uniform magnetic field to enable image encoding



$$B = B_0 + G_x X$$

$$B = B_0 + G_y Y$$

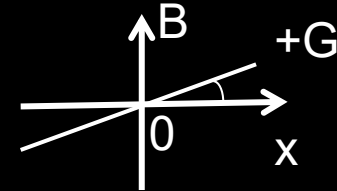
$$B = B_0 + G_z Z$$

$$B = B_0 + G_x X + G_y Y + G_z Z$$

Enables spatial encoding (i.e. imaging) in any arbitrary plane, without any moving parts

Overview: The MRI system

Magnetic Field Gradient subsystem (driving current $\sim 300\text{A}$)



Gradient strength

- Measured in milliTesla/meter (mT/m)
- Stronger gradients-faster data acquisition (DWI), higher resolution
- Amplitudes $< 5\%$ of the main magnetic field (10 to 80 mT/m....)

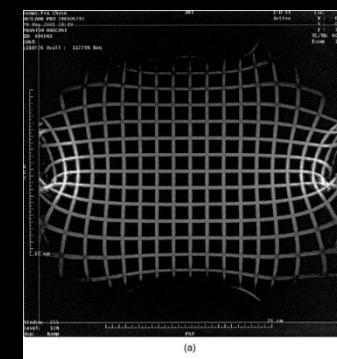
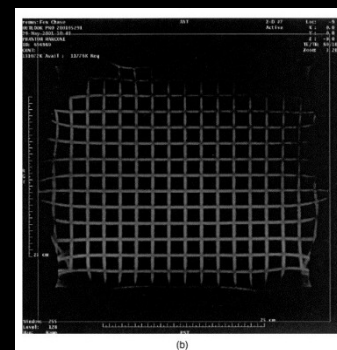
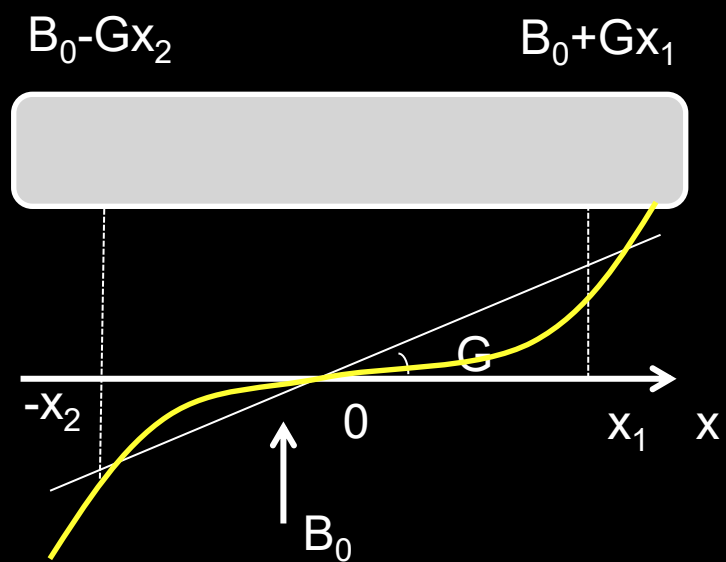
Gradient Slew rate – Rate of switching gradient polarity (mT/m/s)

- Achieved by manipulating the currents through the gradient coils
- Higher slew rate – Faster imaging, (possibility of nerve stimulation), louder accoustic noise, Eddy currents

Eddy currents – Faraday's law of induction

- Parasitic currents created by the changing magnetic field, which counteract the magnetic field change
- Lead to changes the actual MFG form, reduce imaging quality (distortion, blur, ghosting artifacts)

Gradient linearity

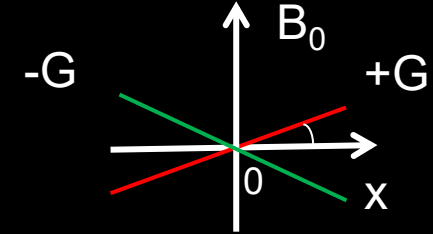
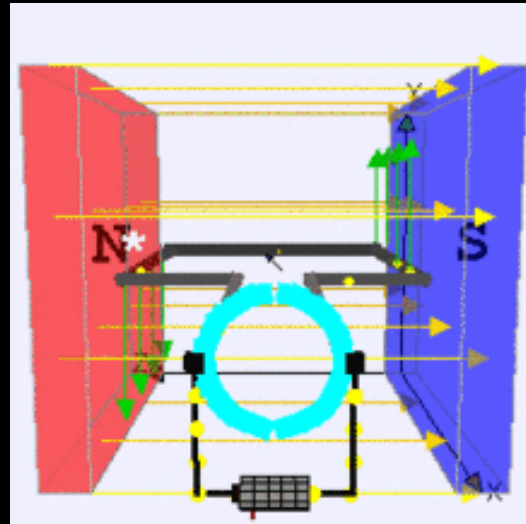


Distortion



Gradient linearity

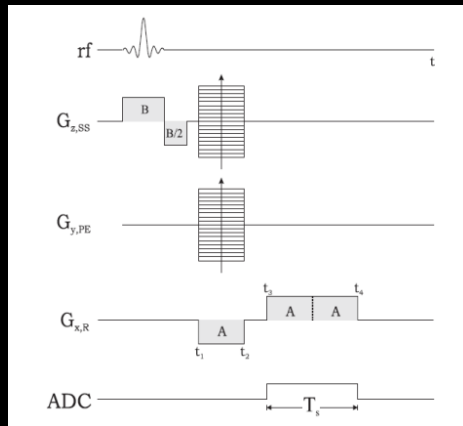
Rapid switching of gradients (by the alternating current that drives these coils) during imaging, leads to dynamic **Lortenz-forces** on these coils – causing them to vibrate



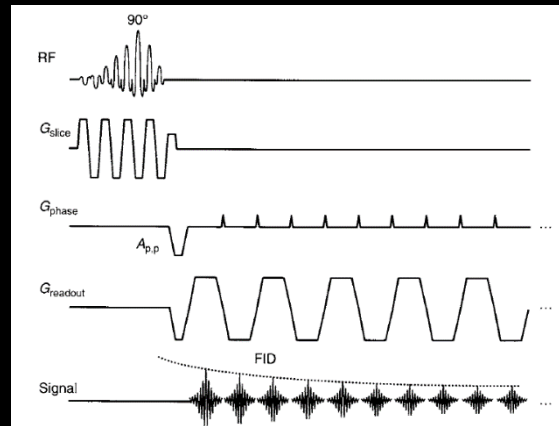
... the same principle that makes this current carrying loop/coil rotate in external magnetic field

Typical sound (vibration) frequencies 20 Hz – 2000 Hz

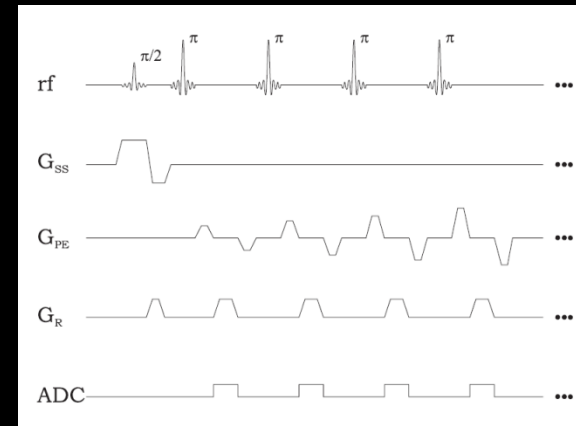
Sounds of MRI



3D GRE



GRE-EPI



RARE (Fast Spin Echo)



Just for fun



Radio Frequency (RF) system (B_1)

RF subsystem

Transmission - To produce appropriate RF pulses needed to generate MR signal

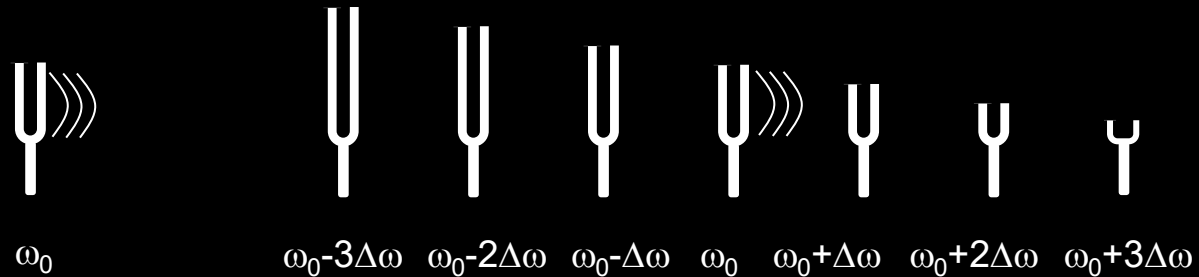
Reception - Capturing, conditioning and processing of the MR signal

Radio Frequency (RF) system (B_1)

RF subsystem

Transmission - To produce appropriate RF pulses needed to generate MR signal

Tuning to
Resonance



Radio Frequency (RF) system (B_1)

RF subsystem

Transmission - To produce appropriate RF pulses needed to generate MR signal

B_0		ω_0
1.0T	—	42.6 MHz
1.5T	—	64 MHz
3.0T	—	128 MHz

Radio Frequency (RF) system (B_1)

RF subsystem

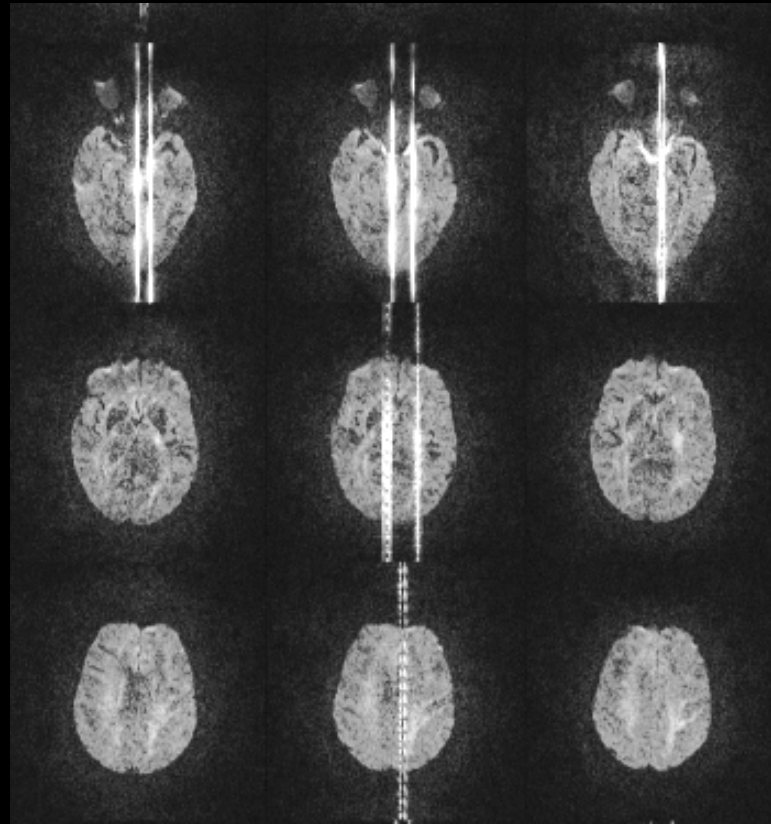
Transmission - To produce appropriate RF pulses needed to generate MR signal

This is close to the frequencies of our radio-stations



Magnet/Scanner Room RF shielding

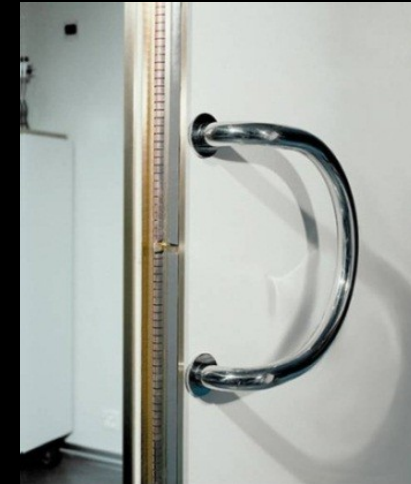
RF Shielding



Magnet/Scanner Room RF shielding

RF Shielding

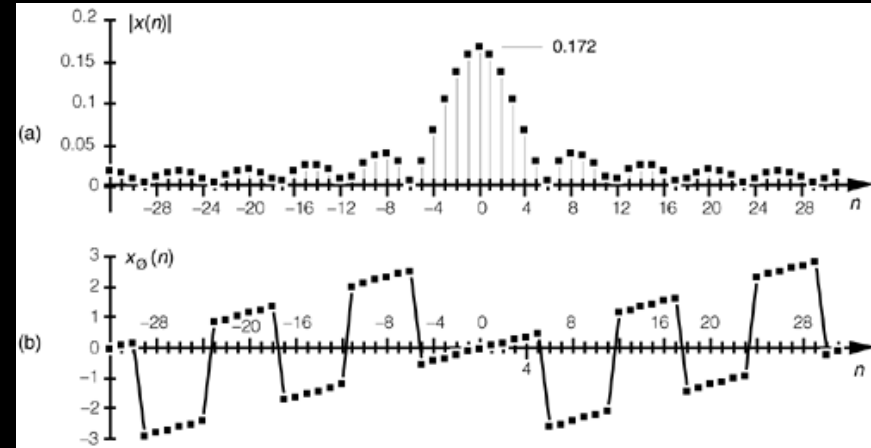
Faraday cage



Steel plates and copper mesh lining with contact sealed doors

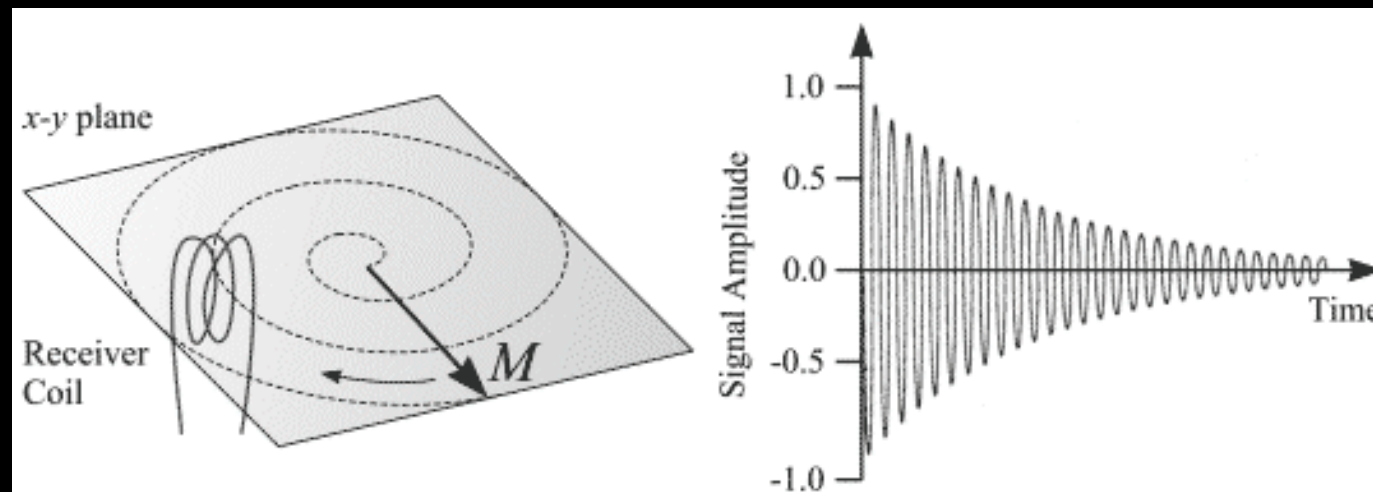
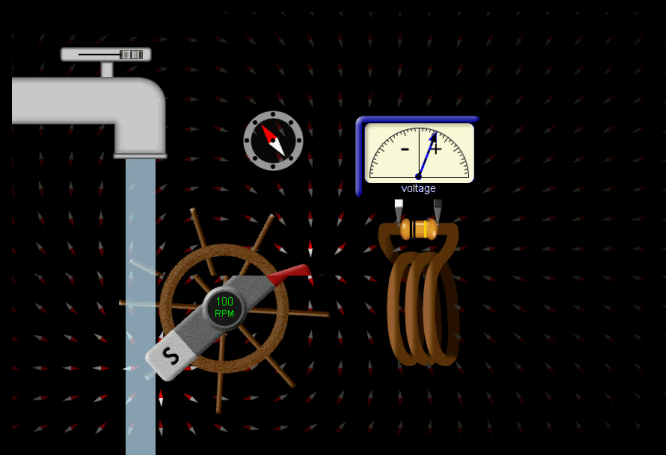
RF subsystem (B_1) - Transmit

- Major components
 - Digital-to-analog converter
 - Amplifier
 - Transmit coil
- Property of RF system
 - RF power
 - Input: proportional to output voltage of amplifier (power ~100W - 50kW)
 - Output: B_1 field power (proportional to B_0^2 ; safety level: SAR<4-8W/kg b.w.)
 - RF Phase
 - RF phase also determines signal outcome



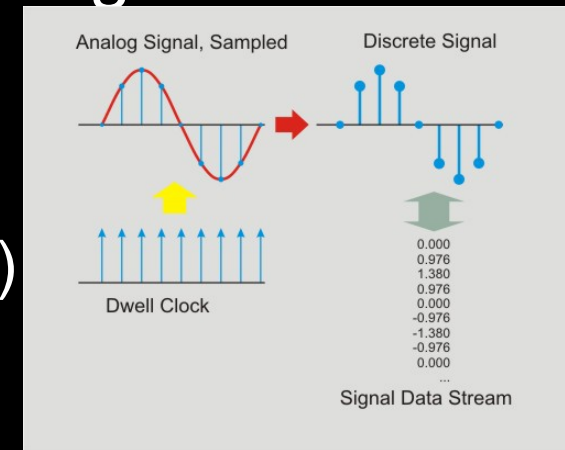
RF subsystem (B_1) - Receive

- Capturing, the MR signal



RF subsystem (B_1) - Receive

- Capturing, conditioning and processing of the MR signal
- Components
 - Receive coil (resonant frequency = ω_0)
 - Pre-amplifier (signal at mV level)
 - Bandpass filter (remove noise)
 - Demodulator (remove B_0 Larmor frequency components)
 - Analog-to-digital converter (ADC)



RF coils

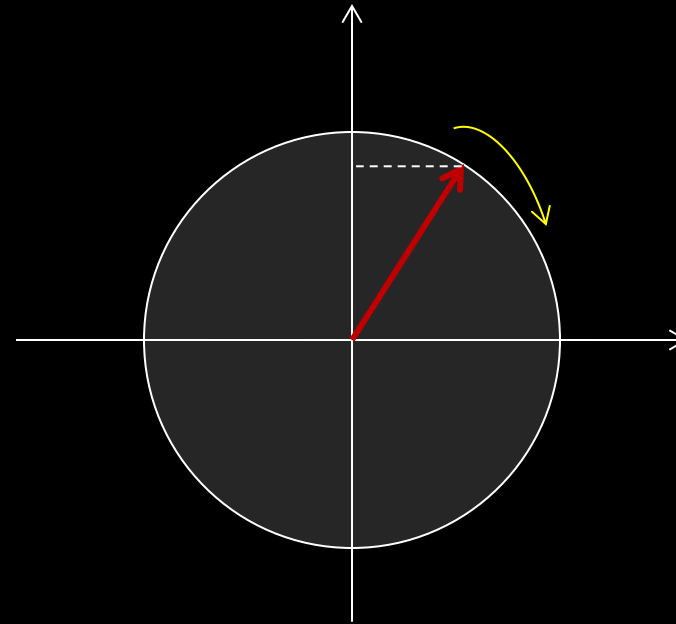
- Transmit (Tx); Receive (Rx); Transmit-and-Receive (Tx/Rx)
- Volume vs. surface
- Quadrature vs. linear vs. phased array
- Body/ head/ neck/ wrist/ knee/ ankle/ shoulder/ breast/ body matrix



RF receiver system

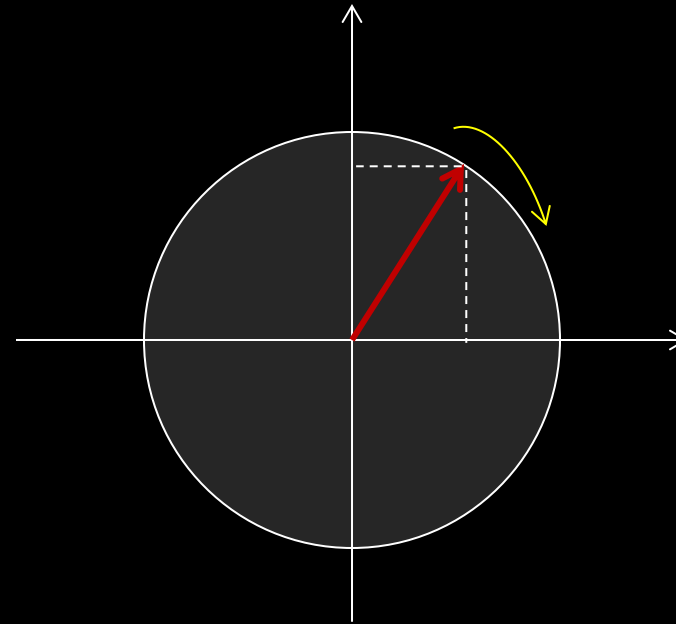
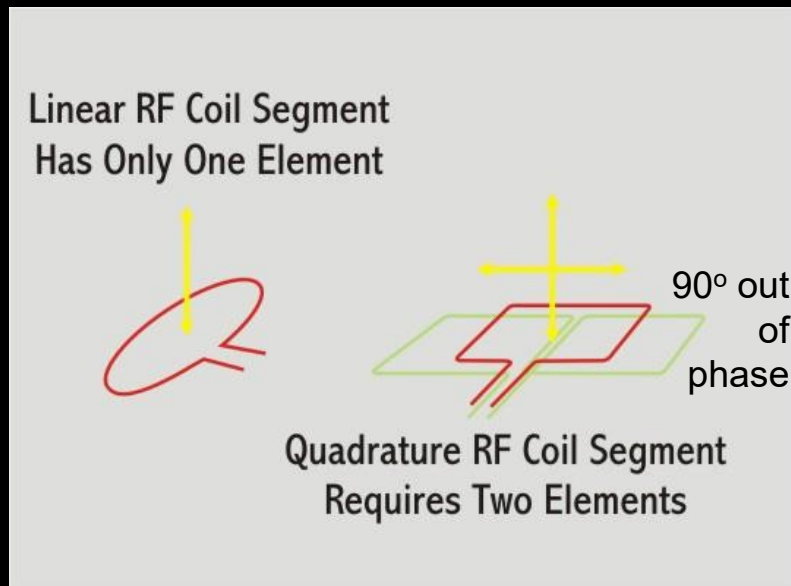
- **Quadrature vs. Linear vs. Phased array coils**
 - Linear coil only captures half of the available signal (linearly polarized)

Linear RF Coil Segment
Has Only One Element



RF receiver system

- **Quadrature vs. Linear vs. Phased array coils**
 - Linear coil only captures half of the available signal (linearly polarized)
 - Quadrature coil contains two perpendicular coil elements, detected signal in each are 90° out of phase (circularly polarized)



Captures both components of the magnetization

RF receiver system

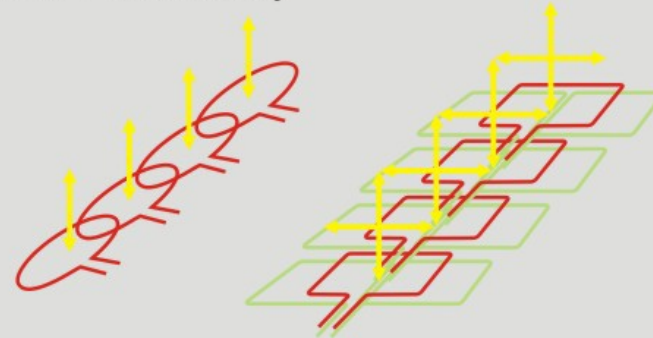
- **Quadrature vs. Linear vs. Phased array coils**
 - Linear coil only captures half of the available signal (linearly polarized)
 - Quadrature coil contains two perpendicular coil elements, detected signal in each are 90° out of phase (circularly polarized)
 - Big boost in SNR with PA coils (theoretically $\sqrt{N}$ folds), or larger imaging FOV

Linear RF Coil Segment
Has Only One Element



Quadrature RF Coil Segment
Requires Two Elements

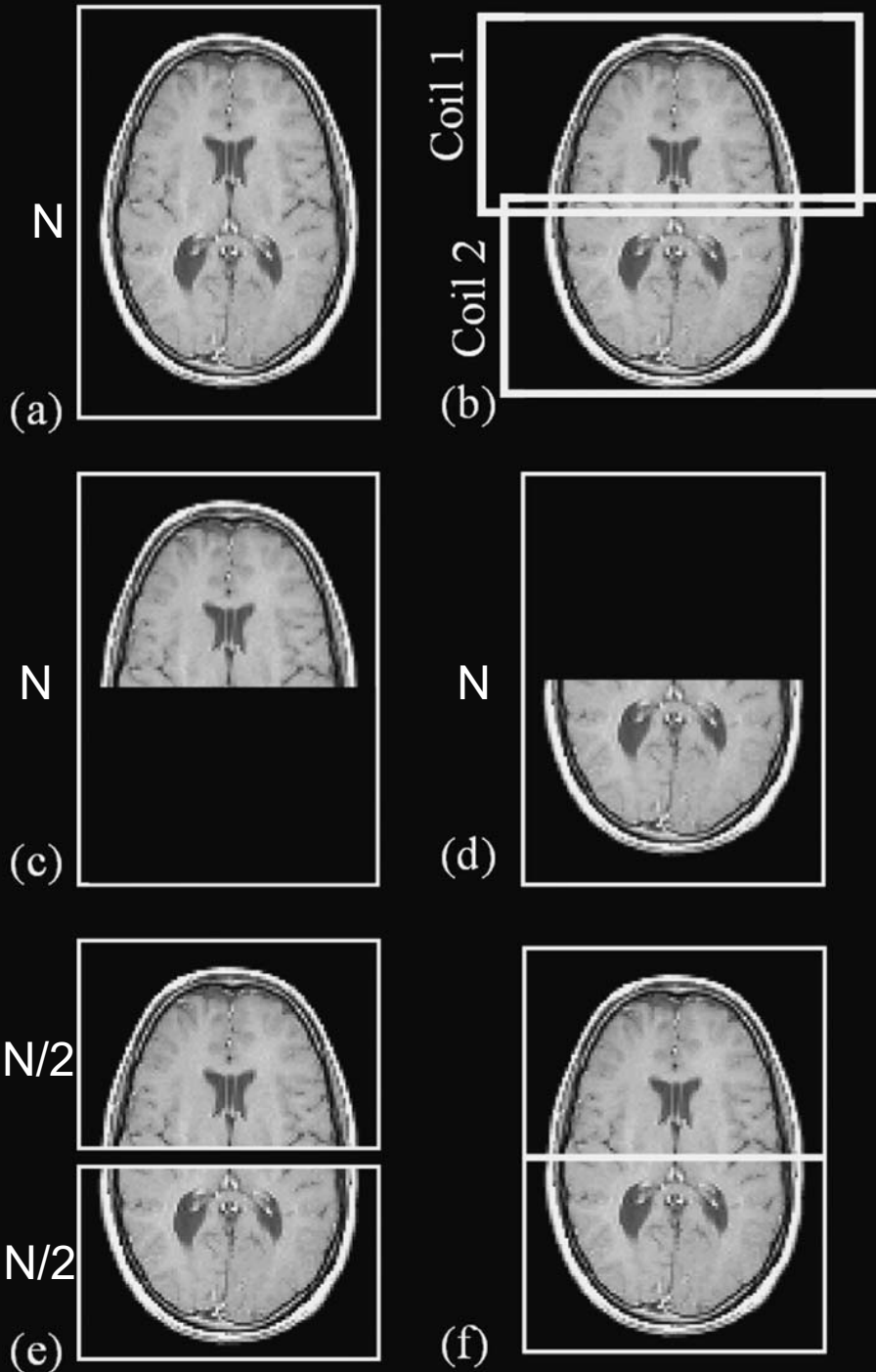
Linear Phased Array



Quadrature Phased Array

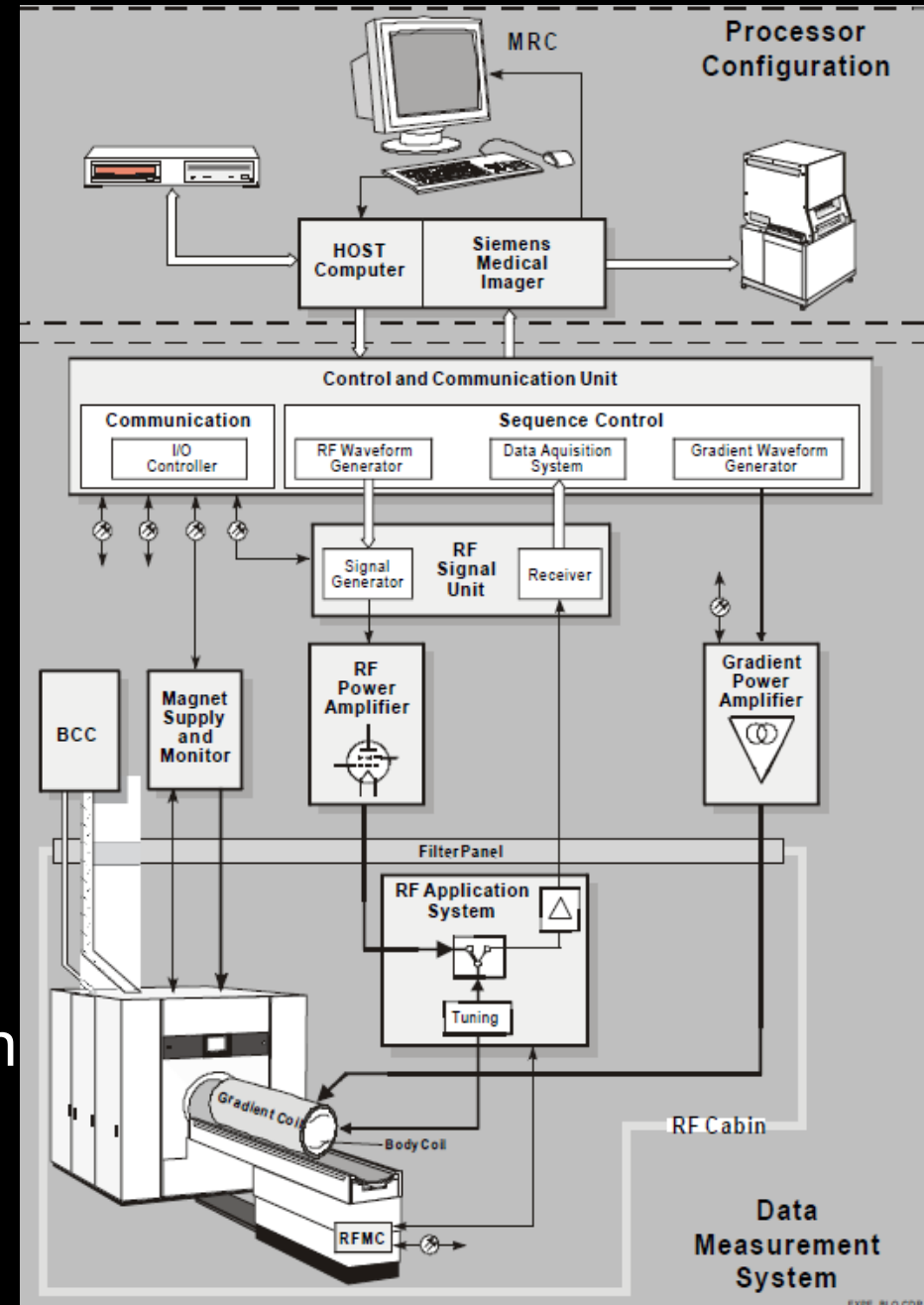
Coils for parallel imaging (pMRI)

- Only phased array coils can be used for pMRI,
- Basic principle
 - Different coil elements cover different spatial regions
 - This can be used to encode position, instead of magnetic field gradients.
 - In practice, a combination of coil localization and gradient encoding is used to reduce total image acquisition time.



Computer sub-system

- System control unit (host)
- Database
- RF transmit/receive system
- Gradient system
- Image reconstruction system
- Monitor system (system/physiology)
- Sequence/reconstruction design system



Summary Points

- A strong, extremely stable (in space and time), magnetic field is required for Magnetic Resonance Imaging
- Both superconducting and non-superconducting magnets are used
Superconducting magnets provide tremendous advantages including
 - better image quality and
 - faster imaging
- Simultaneously controlled gradient coils for X, Y, and Z spatial directions
 - enable selection of imaging plane in any arbitrary orientation
- RF coils – Required for generating and receiving MRI signals
 - Resonant frequency (ω_0) associated with field strength (B_0)
 - Multichannel RF coils – allow time saving through parallel imaging

Key words

Magnetic field homogeneity (inhomogeneity)

Magnetic field gradients – Linear - Uniformity

Radio frequency coils – Transmitter and receiver

Resonance

Quadrature detection

RF leak vs. shielding

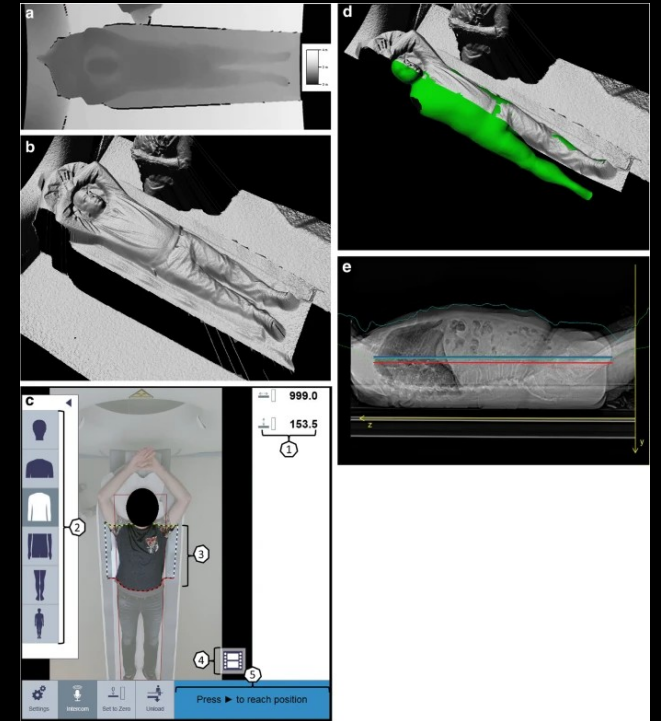
MRI procedure video



MR Safety video

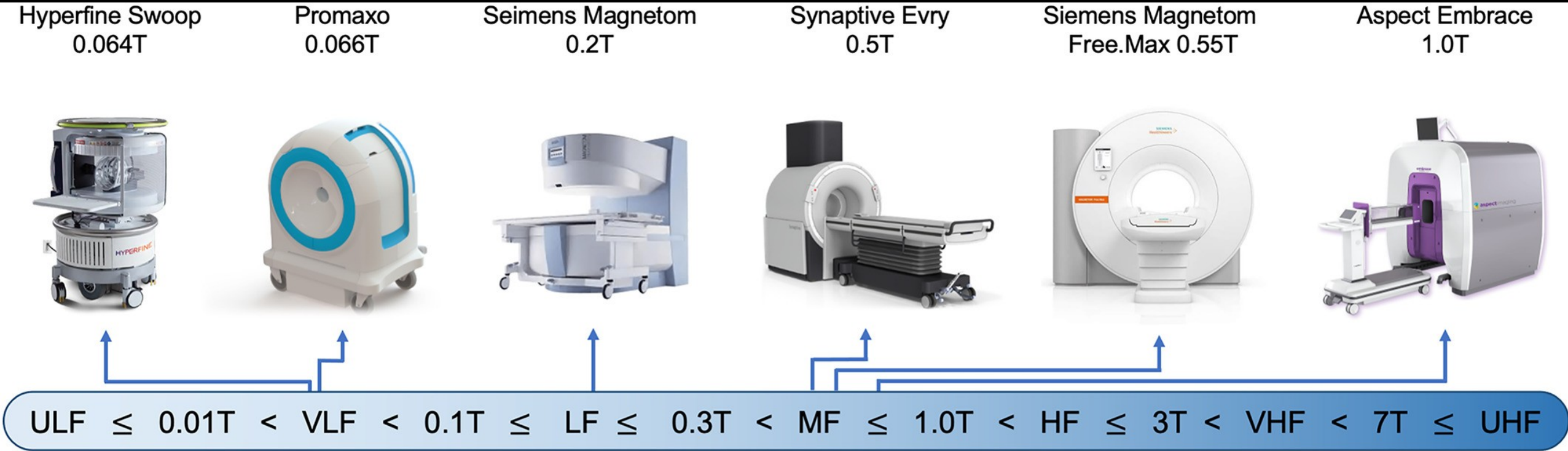
Active areas of research

- 5 min MR exam
- Biomarker development
- Autonomous MR – auto positioning, planning etc.
- Auto tissue segmentation/triaging
- Computer-aided diagnosis (not strictly MRI res.)
- Accessible MR technologies – low-cost or purpose-built scanners
- Point of care MR
- ...



Active areas of research

LOW field MRI is on the resurgence today taking advantage of the recent advances in the hardware and AI based reconstruction methods



Active areas of research

Acquisition:

- Biomarker development
- Novel contrast mech.
- Quiet imaging (Silent MR)
- MR fingerprinting
- High-speed imaging
- Organ motion tracking
- ...

Coils:

- Better coils for imaging different body parts
- Improved patient sensing through radar tech
- ...

Reconstruction:

- Recon/ high speed imag.
- Motion-robust imaging
- Quantitative mapping
- ...

Workflow/pt. comfort

- Auto positioning
- Auto planning
- ...

Applied MR

- fMRI for studying brain
- DTI
- Quantitative MR
- ...

There is active research in each sub-systems of MR and where-ever applicable, AI is being employed